

EKF validation — Summary & Equations

Duration: 34999.16 s $dt \approx 1.0000$ s $n = 35000$

SOC: RMSE=0.243860, MAE=0.185798, Max|err|=0.728614 @ 34385.2 s, Bias=0.168538

Innovation: RMSE=0.030662 V, Bias=-0.013723 V

EKF equations

$$x_{k|k-1} = f(x_{k-1}, u_k) + w_k$$

$$P_{k|k-1} = A_k P_{k-1} A_k^\top + Q_k$$

$$K_k = P_{k|k-1} H_k^\top (H_k P_{k|k-1} H_k^\top + R_k)^{-1}$$

$$x_{k|k} = x_{k|k-1} + K_k (z_k - h(x_{k|k-1}, u_k))$$

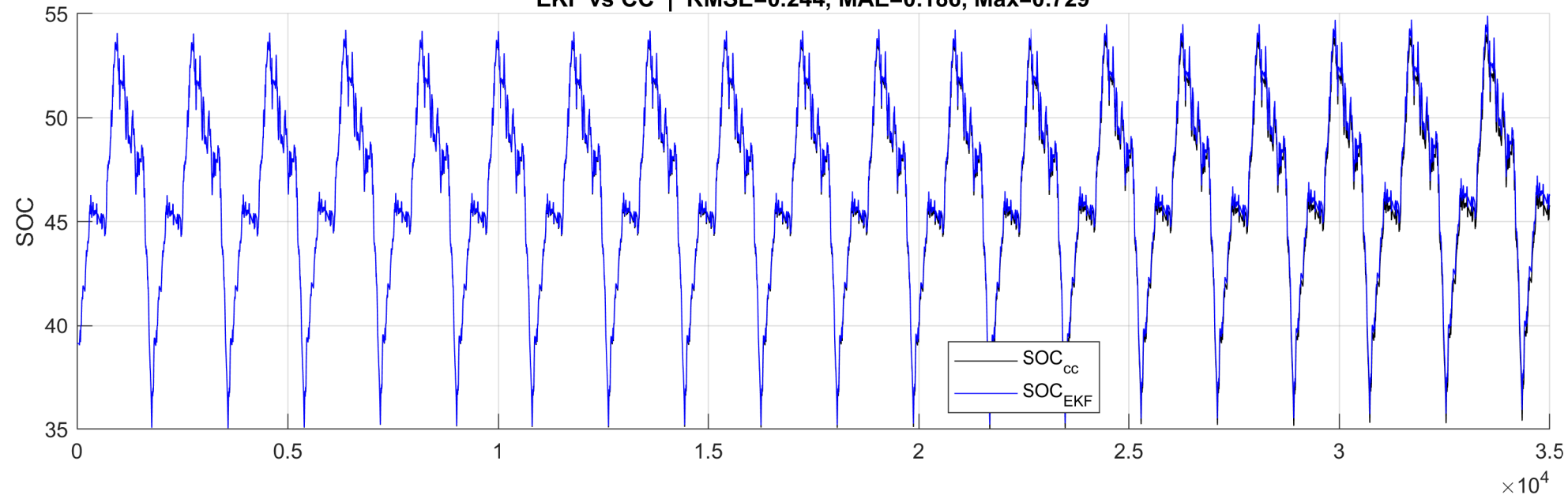
$$P_{k|k} = (I - K_k H_k) P_{k|k-1}$$

$$\text{Innovation } \nu_k = z_k - h(x_{k|k-1}, u_k)$$

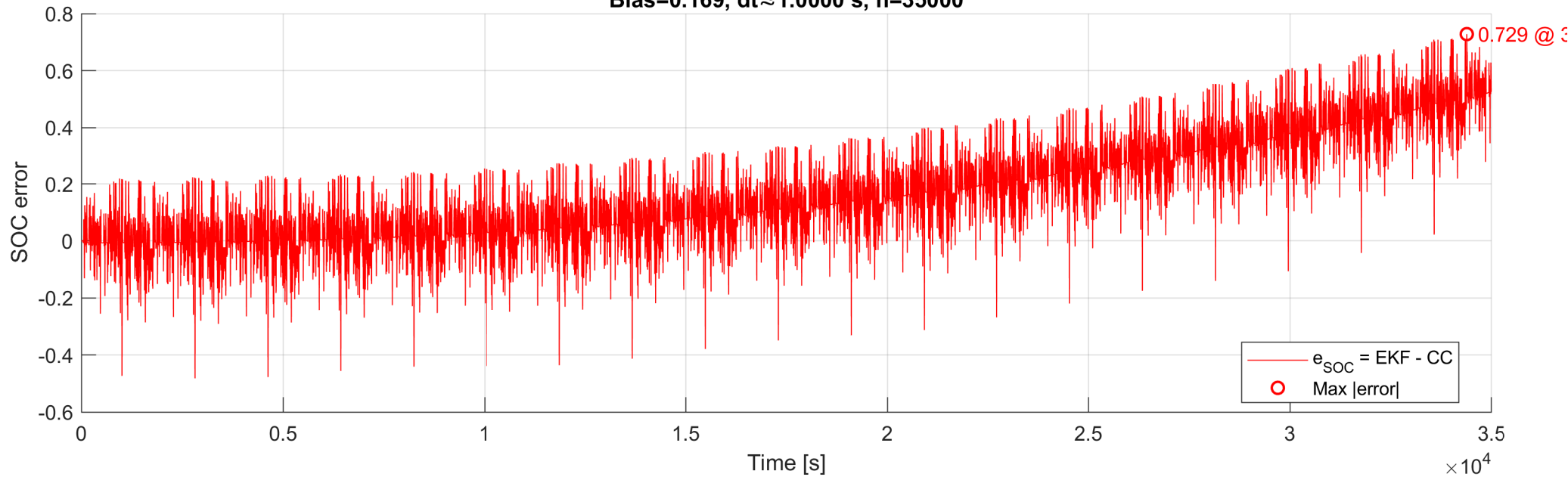
Signals present — Innov: yes K_{out} : yes P_{out} : yes V_{meas} : yes

EKF validation — SOC

EKF vs CC | RMSE=0.244, MAE=0.186, Max=0.729



Bias=0.169, dt $\approx$ 1.0000 s, n=35000



EKF validation — Innovation & Gain

Innovation | RMSE=0.031 V, Bias=-0.014 V

